

Project 1 Week 1 – Design Thinking

Wearable Technology Studio

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Design Thinking - Need Finding

- ***Need finding*** is discovering what an individual/population needs and bridging the gap between their ***goals/values*** and ***experience***.
- It ends with a ***need statement*** which is of the form:

“[Client/Population] needs [a type of solution] which [conforms to design constraints] in order to [further goal/provide value].”

Example: “People with carpal tunnel syndrome need a treatment option that is effective, easy to use, and pleasurable to alleviate symptoms and avoid surgery.”

Design Thinking - Brainstorming

- There are many brainstorming techniques
 - Empathy, Interviewing, “Painstorming”
 - Worst Idea
 - Mind-mapping
 - Thinking Hats/Roleplaying
- Goals: Variety, Exploration, Divergent Thinking

Project 1 – “Light-up Costume”

Wearable Technology frequently includes lights. Your wearable technology company will be debuting a new design concept aimed at costumes/uniforms/future fashion. Think about how to create value for your customers.

Constraints

Your wearable product must adhere to the following constraints:

- 1) Wearable textile component
- 2) Contain LEDs and/or other lights
- 3) Contain a 3D printed component
- 4) Be functional and provide value to users

Project 1 Agenda

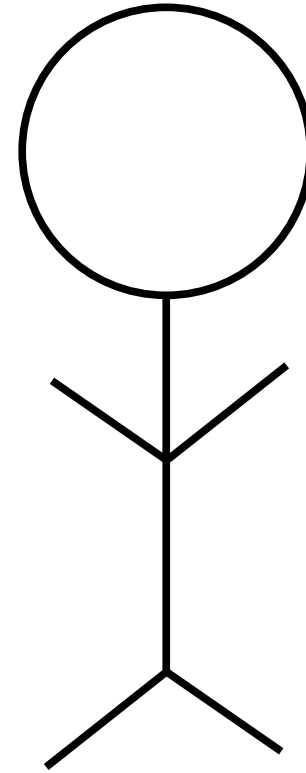
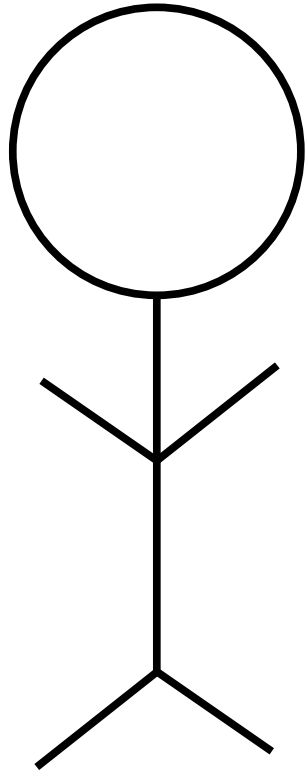
Schedule

- **First week (1): Design Concept for “Light-up Costume”**
- Second week (2): First Textile Prototype
- Third week (3): LED/Battery/Wiring Design
- Fourth Week (4): 3D Printed Component
- Final Design Presentation (5)

Purpose:

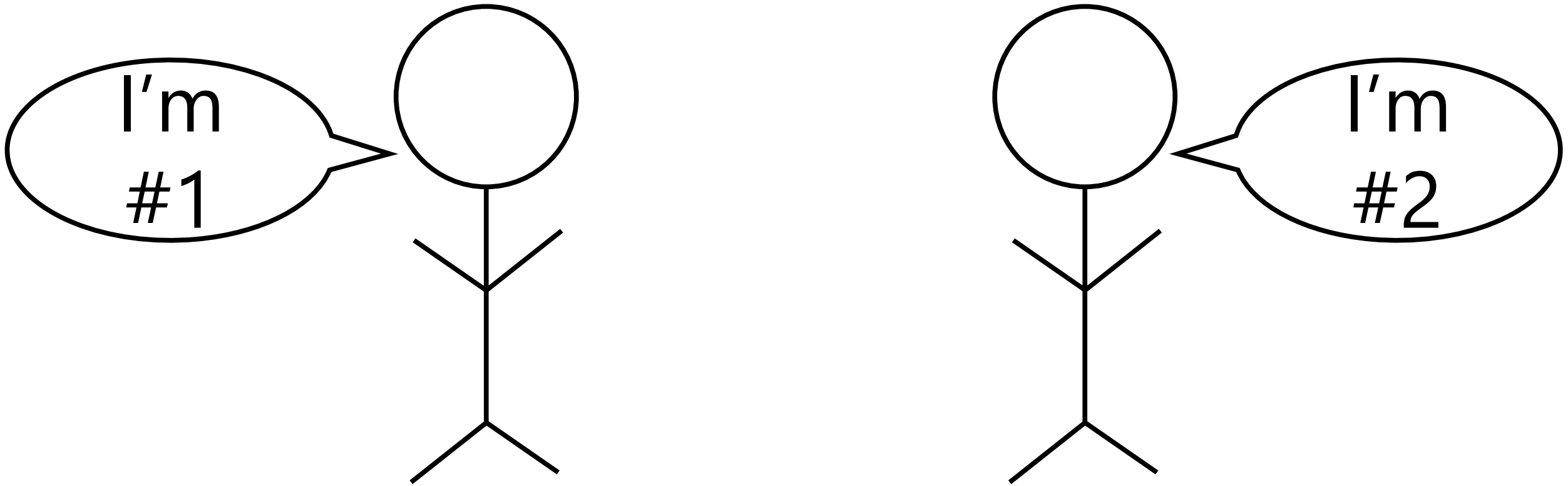
- To design a wearable product
- To brainstorm design concepts and refine them
- Finalize a design concept aimed to fulfill a need

Zero Fidelity Prototyping



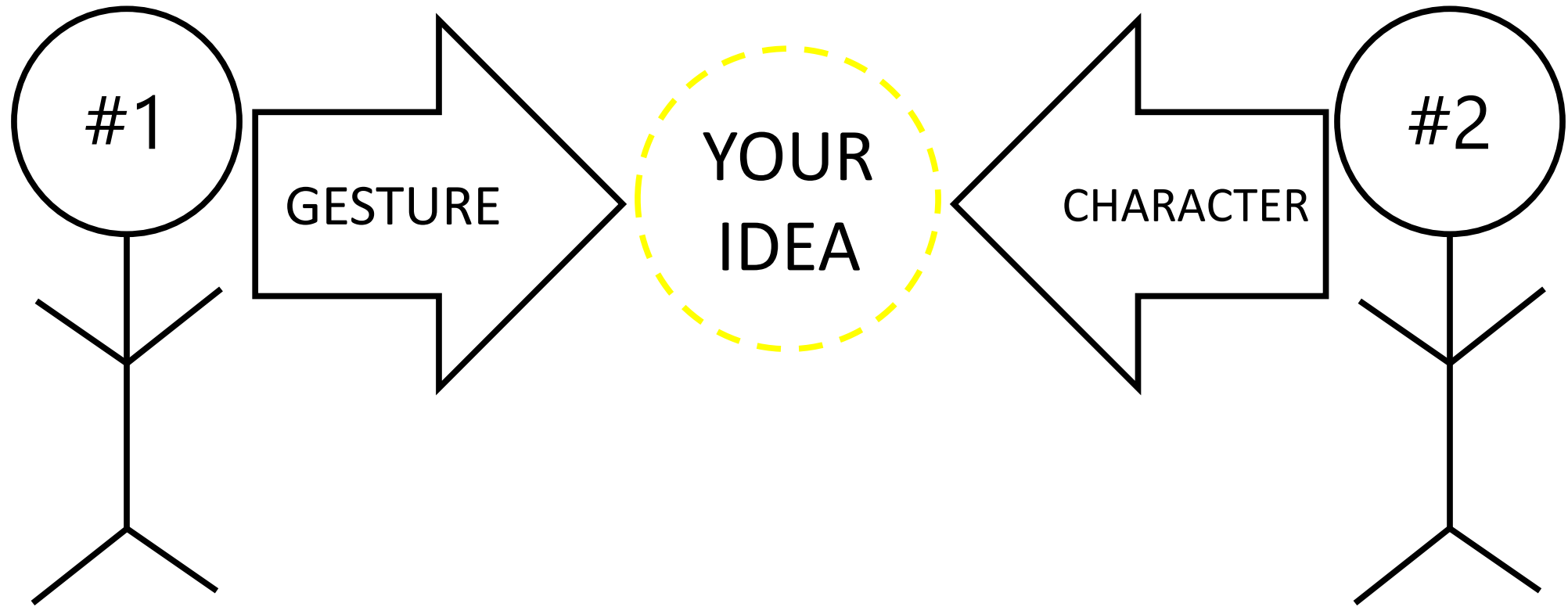
FIND A PARTNER

Zero Fidelity Prototyping



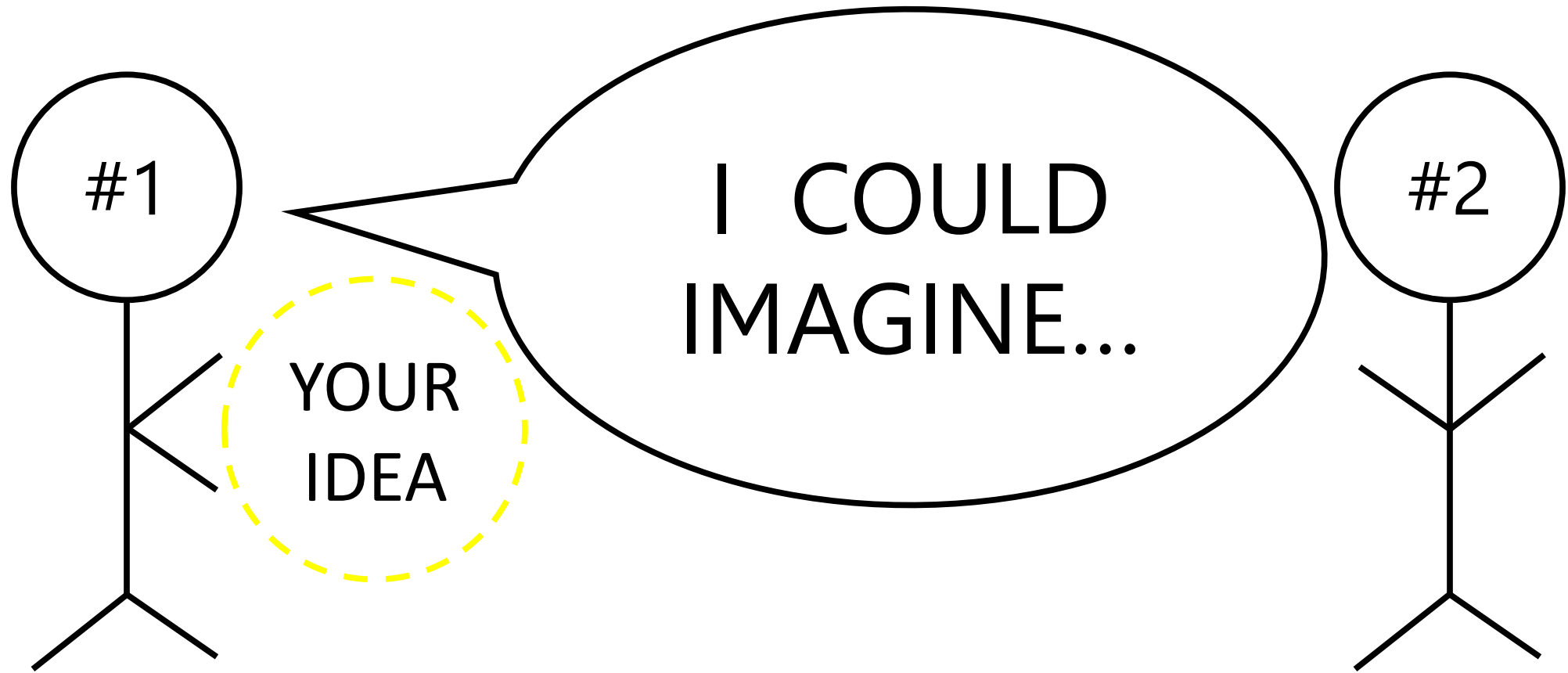
ASSIGN ROLES

Zero Fidelity Prototyping



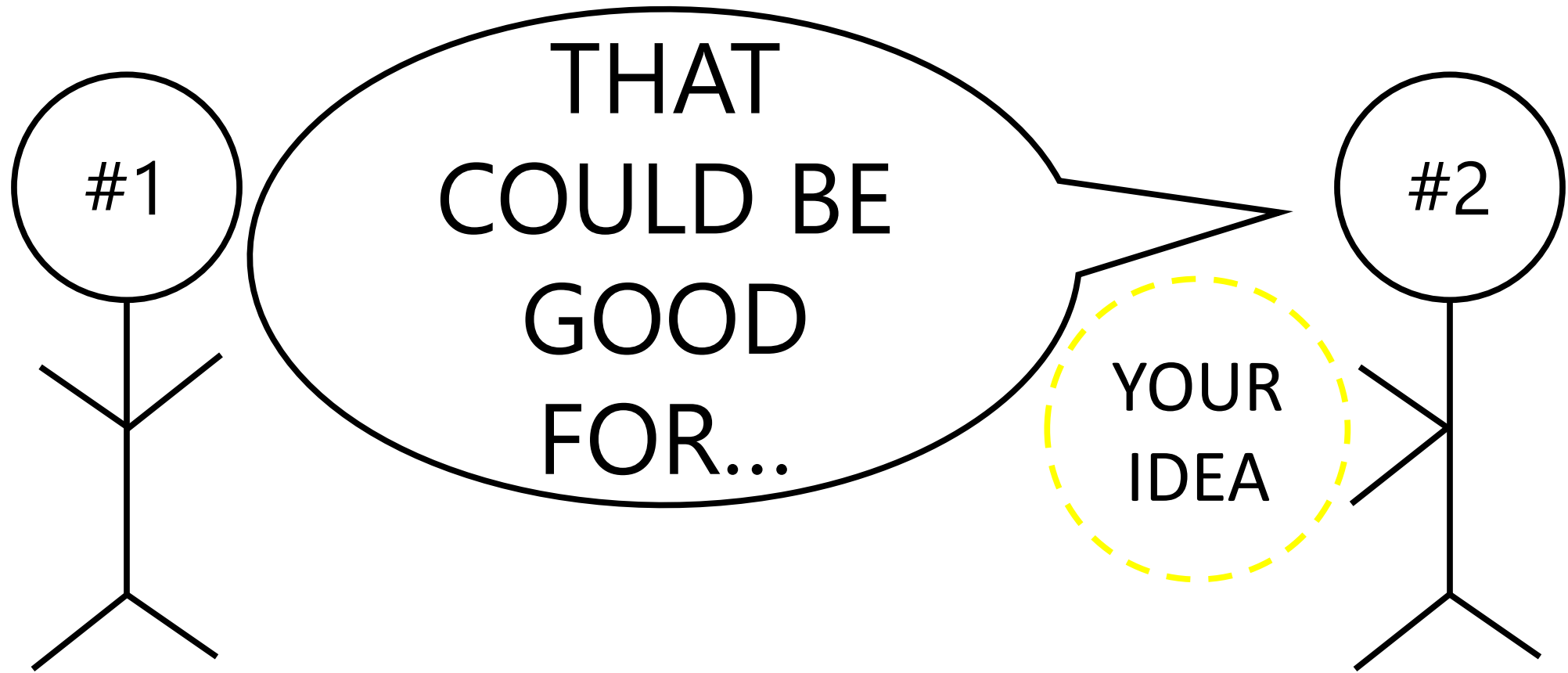
BISOCIATION

Zero Fidelity Prototyping



BRAINSTORMING

Zero Fidelity Prototyping



NEED FINDING

ZFP → First Draft Design Concept

- What is it? What does it do?
- Who is it for? Why would they want it? (Need Finding)
- What will it look like?
- What will it be made of
- Draw Sketches!

SCAMPER for Brainstorming

- SCAMPER is a technique for “Turning All The Knobs” on a design concept
- Great for avoiding “first idea syndrome”, exploring all possibilities
- Best done as part of a longer brainstorming session

AI for Brainstorming

- AI is good at generating many plausible ideas *quickly*
- Best Practice – Engineering Context, Not Prompt
- Best Practice – Sketches → Image Generation for Imagination
- Upload the SCAMPER Worksheet into an AI (Gemini 2.5 Pro) along with a description of your first draft design concept and sketches
- PROMPT: I am designing a light-up costume for a Wearable Technology Design Studio course. Here is my design concept so far. Follow the SCAMPER prompts to brainstorm many variations on this design concept.

After Brainstorming, Selection

- Brainstorming should result in *a lot* of ideas
- Evaluation requires selection criteria (objectives and constraints)
- Start by screening out insufficient designs
- Make a design matrix, but don't abandon intuition or blindly follow the matrix

Criteria	Steel	Aluminum	Plastic
Cost	1	0.5	0.5
Strength	1	0.5	0
Weight	0	0.5	1
Total	2	1.5	1.5

Project 1 – “Pinup” Report Out

Share the following:

- Your first design concept (components, function, etc.)
- Some of your brainstorming variations
- Who is it for? Why would they want it?

Project 1 – Homework

Upload the following to the Discussion Board:

- Your first draft design concept, including sketches
- The output from your AI prompt and a link to the shared chat
- Your finalized design concept with sketches (also any AI-generated images)
- Answers to the posted questions